Ans) A high power electric vehicle drive system is a significant source of electromagnetic interference EMI, primarily due to the rapid switching of ~~an~~ electromagnetic interference (EMI) primirily due to rapid switching of power semiconductor in the motor inverter. This generate high dv/dt and di/dt transient which couple as noise into sensitive electronics control system, leading to malfunction. Enhancing electromagnetic compactibility requires a comprehensive, multi-layer startegy that ~~add~~ address a ~~comprehensive multi layers~~ ~~st~~ three fundamental elements of any EMI probl. the source where the noise is generated, the path (how the noise travels and the victim.

The EMI Imperative in electric vehicle

The co-loc of high power switching circuits and highly sensitive electronics control unit creates a challenging electromagnetic environment. Drive system EMI typically manifest as.

1) Conducts emmission: Noise currents travelling along power lines and ground path

2) Radiated Emmission: Electromagnetic waves propagition through space

1) Migration at the source: Reducing EMI Generation

The most effective step is to prevent the generation of noise at it's origin, the power converter

A) Power Semiconductor Switching control.

- Slow rate optimization: Controlling the gate drive volt and current of the power ~~sw~~ switching to manage the turn on and turn off speeds. A slightly reduced slew rate significantly lower the high-freq. harmonic content often reducing conducted EMI over 10dB, without a major drop in efficiency.
- Soft switching technique: Implementing advanced topologies like zero voltage switching or zero-current switching. By ensuring the switch turn on or off. When the volt or current across its ~~a~~ zero, transient spikes ~~a~~ are minimized ~~substan~~ substantially reducing switching losses and high frequency emmission
- Pulse width modulation: Empolying tech. Such as random PWM or chaotic pwm, these techniques spread the noise energy across a wider range of freqency instread of connecdrating it at specific harmonic peaks, making it easier to meet regulating limits.

B) optimisezed power module and Busbar Design

- low inductance packaging: Utilizing advanced power module packaging and laminated bus bars. b/w en the DC line capacitor and the inverter switching. stemminated busbars minimize the parasitic loop inductance of the DC line which directly reduces the magnetic of volt overshoot and high freq ringing

2) ~~Miga~~ Mitigation: via path: Interrupting the EMI Coupling

A) filtering for conducted emmissions

- Common mode and differential mode filters: A complete EMI filter at the I/p of the inverter consist of both DM and CM elements.

- Filter placement - filter must the located precisely at the point of entry of exit of a shield enclosure ~~to~~ to ensure that all conducted noise is blocked before it can propagate into the cable harness, which acts as a larger antenna

B) shielding and surrounding for radiating emmission

- Metallic enclosure: Encassing the high power converter and motors in robust metallic closures. these act as faraday cagle to contain radiated emmission. Conducting gastting is often used low-impedence contact accross enclosure mating surface

- shielding cable and ~~ter~~termination: Using braided or foil shield cable for both high volt power lines and critical low voltage signal hardness. the shield provide a low impedence return path for noise current and confines the electromagnetic field

c) Cable Routing and Separation

• Separation of domains: Maintain max physical dist b/w the noise source cable and the victims signal cables.

• Orthogonal crossing: when a noise source ~~cable~~ cable must cross a sensitive cable, they ~~so~~ should cross at a 90° angle to minimize inductive and capacitive coupling

• Twisted pair wiring: Utilising twisted pairs cables for all differential signal lines. the twisted cancels ~~the~~ the induced magnetic noise by ensuring the noise is picked up equally on both conductors, allowing the differential ~~the~~ receiver to reject the common mode of noise

3) Migration at victim: Hardening Susceptibility

1) PCB design techniques

• Solid ground planes: Using multi layer PCB's with continious dedicated solid ground and power planes. This minimize ground impedance reduces loop areas for the high freq return current, and provides intrinsic shielding b/w layer.

• Decoupling capacitor: Placing a sufficient density of high freq decoupling capacitors as close as possible to the power pins of every IC and microcontroller

• Trace Geometry : keeping high speed signal traces short and away from board d edges and mouting them over an unitrupted ground plane to control impedence and minimize coupling

B) Transient Immunity

• Transcient voltage suppressors : Implementing TVs diodes zener diodes and varistors at all EV input /output ports to clip high energy transcient that could damage the sensitive internal ICs.

• ferrite loads : Inserting ferrite loads in series on signal and power lines at the load entry to act as a high frequency ressistor. dampning noise currents before they reach the sensitive IC.

c) Software and firmware.

• Digital filtering : Implement noise suppression algorithm, such as moving avg filter in the ECU software. to condition sensor and prevent ~~~~ control action that might be triggered by EMI induced volt. glitches

• timer : Used to quickly detect and safely recover from EMI induced processor malfunction

By integrating these measures across the source, path, victim EV manufacturing can efficiently manage the sourse EMI challenges.

Q2) Ans, Reliability against external disturbance like solar radia begins with a robust communication link and signal processing design focusing on maintaining the signal to noise ratio and utilising the electromagnetic spectrum effectively.

1. Robust freq. management and spectrum use

- Frequency band selection

- Higher frequency : Communication at higher frequency is less susceptible to ~~rain fadement~~ ionspheric scintillation. while higher bands are susceptible to rain fade on earth, the effect of ionspheric interference is proportional to the inverse square of the frequency

- Solar Radio Noise : However, the sun's radio noise generally increase with frequency which is a consideration for predicting and managing sun outrages

2. Advanced waveform and coding techniques

- Spread Spectrum techniques :- Methods like direct sequence spread spectrum or frequency hopping spread the signal's energy over a much wider bandwidth than the information itself. This dramatically increase the system's immunity to narrow band EMI as the interfarence's energy is only a small portion of the total spread signal's bandwidth

• Higher order forward error correction :- utilizing powerful coding ~~sc~~ scheme allows the receivers to correct a large number bit errors. introduced by transient noise and EMI without needing retransmission. This effectively lowers the required operational SNR, making the communication more reliable during periods of high noise.

• Adaptive coding and modulation :- This technique allows the satellite and ground station to dynamically switch to a more robust but lower -data - rate, modulation and coding scheme when line conditions degrade due to an increase in EMI ensuring continious service, even if at a reduced throughout ~~to~~ survive and function in the harsh environment

3. Electromagnetic compactibility and shielding

• Physical shielding :- Sensitive electronics are enclosed in ~~fara~~ faraday cages to reflect both incoming and outgoing electromagnetic fields. materials like aluminum or composites with conductive coating are used.

• EMI baskets and seals are employed at all seams, joints, and ~~d~~ endless panels to ensure the enclosure and maints the full shielding, efficiceness and prevents electromagnetic leakage

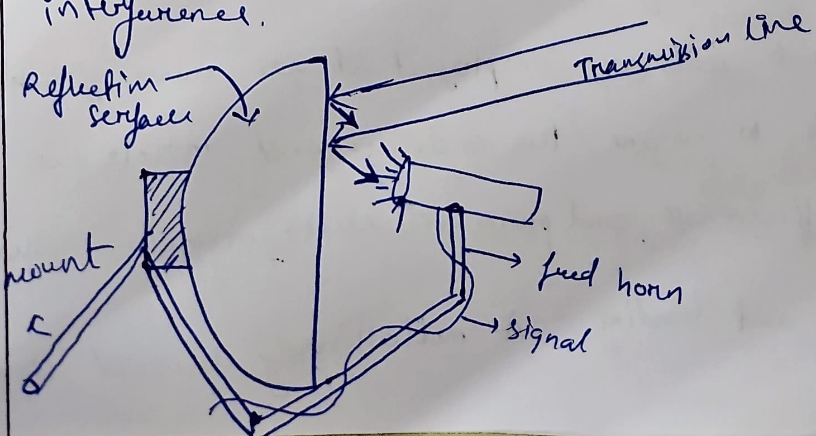
• ~~Gro~~ Grounding and bonding :- A meticulously designed signale point grounding scheme for the satellite's chassis preven the information of ground loops, which can act as unintentional antennas and introduced substanitial conducted EMI.

- Filtering and decoupling :- Installation of low pass filter on all signal and power lines entering ~~compo~~ compartment to block high frequency noise

- Use of decoupling capacitors placed physically close to the powerlines of ICs to suppress instanteneous voltage spikes and transient noise generated by logic switching within the circuits.

4) Adaptive antenna and Beam forming techniques

- High Gain narrow Beamwidth antenna :- Using high gain antennas narrows the antenna's beam width physically restricting the angular are the form which noise distributes noise source on adjacent satellite interferences.



• Adaptive beaming :- In presence of a strong localised interference source the satellite can use its phased array or multi feed antenna to electrically steer in the direction of the interfarence. significaltly alternating the unwanted signal while maintaining communication with the intented ground station

5) Space weather prediction and proactive operation

• Maneuvering and system fating : upon a severe space weather warning non essential system may temporary shutdown into a sayl mode to reduce the risk of power surges

• Redundancy and diversity : for constallation, using ICs provies a higher immune reductant path for data transfer that by pass the potentially turbulent earth space link during severe ionssphric weather.

•

* In summery reliable satellite communicatios in the presence of space EMI and solar radiation requires a holistic design that combines links robustness hardware resiliance and intelligence and adaptive operational startegies.

3)
4)

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Further questions are in form of simulation, so the images of simulated software are attached further.

Q.3) A lossless 30 m long transmission line with Zo=50Ω is established between two ground stations which operate at 2 MHz. The line is terminated with a load ZL = 60+j40Ω. If u = 0.6c on the line. Write a Scilab code to plot the reflection coefficient (Γ), standing wave ratio (S) and input impedance in smith chart.

Source Code:

```
// Scilab script for Q3
// Plots: |Gamma| vs distance, SWR vs distance, and Smith-chart (Gamma plane)

clc;
clear;
close;

// Given
Z0 = 50;            // ohms
ZL = 60 + %i*40;        // load
f = 2e6;            // Hz
c = 3e8;            // m/s
u = 0.6 * c;         // propagation velocity
L = 30;             // line length in meters

// Derived
lambda = u / f;
beta = 2 * %pi / lambda;

// Reflection coefficient at load
Gamma_load = (ZL - Z0) / (ZL + Z0);
Gamma_load_mag = abs(Gamma_load);
```

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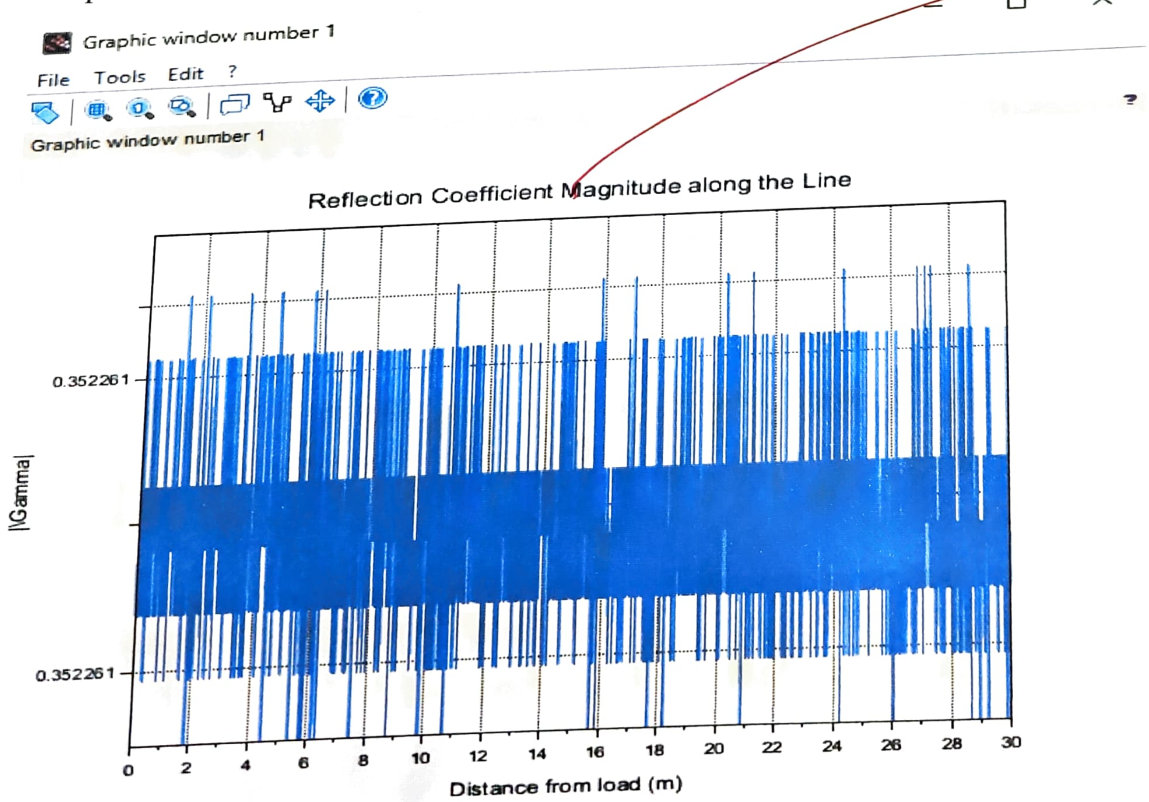
```
Gamma_load_phase_deg = atan(imag(Gamma_load)/real(Gamma_load)) * 180
/ %pi; // approximate angle
SWR_load = (1 + Gamma_load_mag) / (1 - Gamma_load_mag);

// Print numeric checks
disp("Gamma at load: " + string(Gamma_load));
disp(" |Gamma| at load: " + string(Gamma_load_mag));
disp(" Phase (deg) approx: " + string(Gamma_load_phase_deg));
disp(" SWR at load: " + string(SWR_load));

// Sample points along line (0 -> L)
d = 0:0.01:L; // meter resolution (adjust if needed)

// Input impedance along line (lossless)
Zin = Z0 * (ZL + %i*Z0 .* tan(beta .* d)) ./ (Z0 + %i*ZL .* tan(beta .* d));

// Reflection coefficient along line
Gamma_d = (Zin - Z0) ./ (Zin + Z0);

// Plot |Gamma| along the line
figure(1);
plot(d, abs(Gamma_d));
xlabel("Distance from load (m)");
ylabel("|\Gamma|");
xtitle("Reflection Coefficient Magnitude along the Line");
xgrid();

// Plot SWR along the line
```

```
SWR_d = (1 + abs(Gamma_d)) ./ (1 - abs(Gamma_d));
figure(2);
plot(d, SWR_d);
xlabel("Distance from load (m)");
ylabel("SWR");
xtitle("Standing Wave Ratio along the Line");
xgrid();

// Smith chart (Gamma plane)
// Unit circle
theta = 0:0.01:2*%pi;
cx = cos(theta);
cy = sin(theta);

// Prepare points for overlay: Gamma trajectory
realG = real(Gamma_d);
imagG = imag(Gamma_d);

// Compute Gamma at input end (d = L) for annotation
Gamma_input = (Zin($) - Z0) / (Zin($) + Z0); // Zin($) = last element
// (Gamma_load already computed for d=0)

// Plot unit circle and Gamma trajectory on same axes
figure(3);
plot(cx, cy, 'k'); // unit circle
// Overlay trajectory and key points in one plot call
plot(cx, cy, realG, imagG, 'r-', real(Gamma_load), imag(Gamma_load), 'bo',
real(Gamma_input), imag(Gamma_input), 'gs');
```

```
xlabel("Re(\Gamma)");
ylabel("Im(\Gamma)");
xtitle("Smith Chart (Reflection Coefficient Plane) - Trajectory of \Gamma(d)");
xgrid();
// keep equal axis scale
a = gca();
a.isoview = "on";
legend(["Unit circle", "Gamma trajectory", "Load (d=0)", "Input (d=30 m)"], 1);

// Optionally annotate numeric values near points
// (Simple text annotations)
xstring(real(Gamma_load)+0.03, imag(Gamma_load), "Load (d=0)");
xstring(real(Gamma_input)+0.03, imag(Gamma_input), "Input (d=30 m)");
```

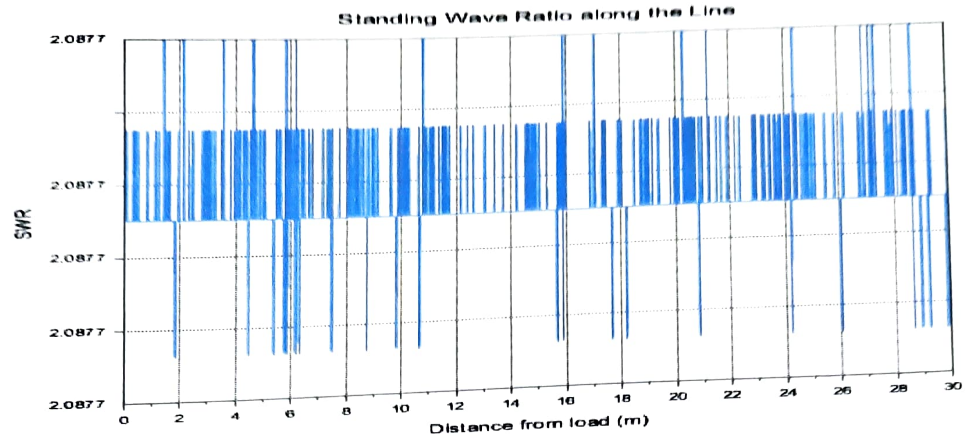
Output:



Graphic window number 2

File Tools Edit ?

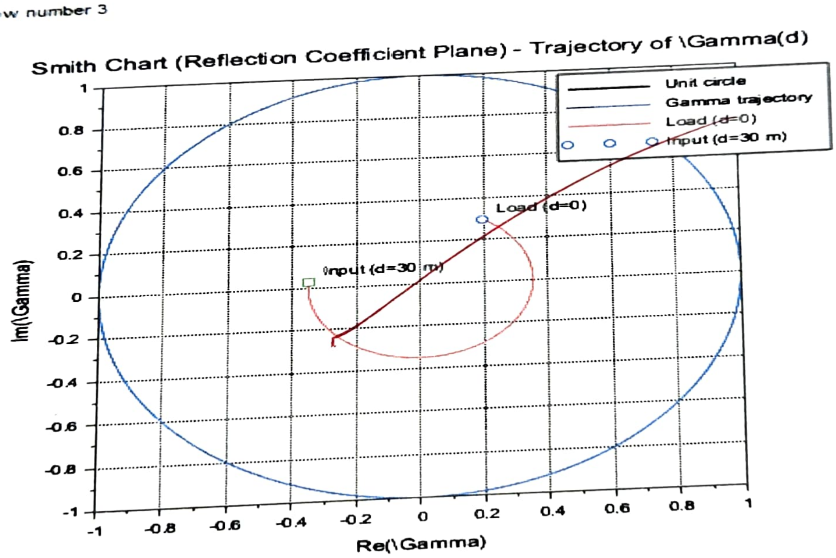
Graphic window number 2



Graphic window number 3

File Tools Edit ?

Graphic window number 3



Scilab 2026.0.0 Console

File Edit Control Applications ?

Scilab 2026.0.0 Console

```
"Gamma at load: 0.1970803+%i*0.2919708"
" |Gamma| at load: 0.3522607"
" Phase (deg) approx: 55.98065"
" SWR at load: 2.0876619"

-->
```

Q.4) In a satellite base station, a load of 100 + j150 Ω is connected to a 75Ω lossless line. Write a Scilab code to plot the Reflection coefficient (Γ), SWR value and input impedance (Zin) at 0.4λ from the load.

Source Code:

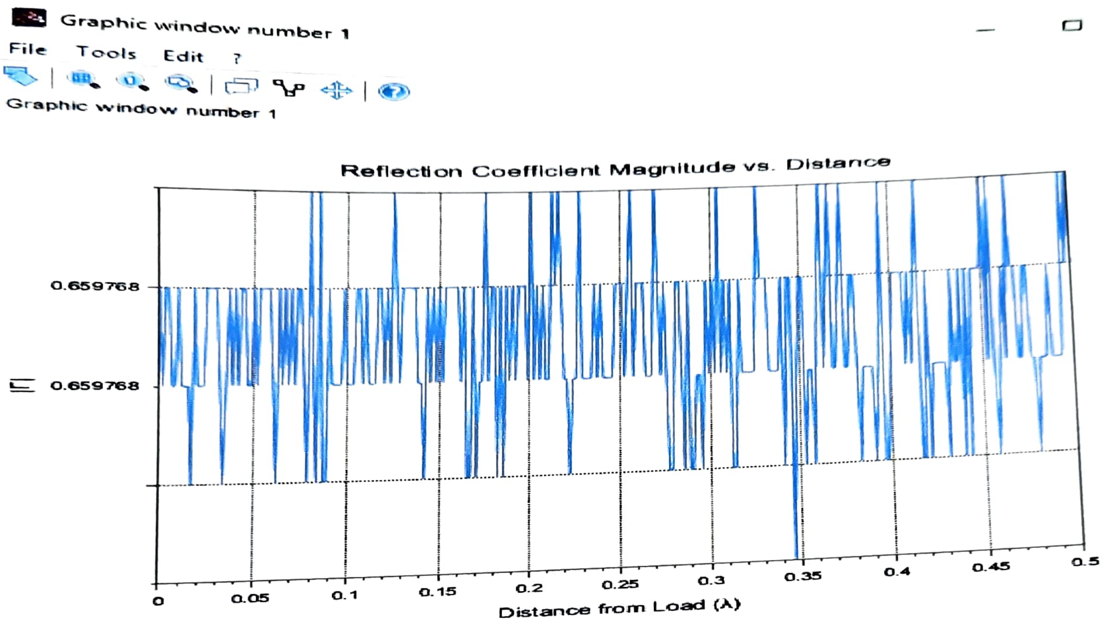
```
// Q4 - Electromagnetic Theory and Interference
// Satellite base station simulation
// Compute and plot Reflection Coefficient, SWR, and Zin at 0.4λ


clc;
clear;
close;


// Given data
Z0 = 75;              // Characteristic impedance (ohms)
ZL = 100 + %i*150;    // Load impedance (ohms)
lambda = 1;           // Normalized wavelength (unit value)
d = 0.4 * lambda;     // Distance from load (in wavelengths)


// Reflection Coefficient at Load
Gamma_L = (ZL - Z0) / (ZL + Z0);
Gamma_mag = abs(Gamma_L);
Gamma_phase_deg = atan(imag(Gamma_L)/real(Gamma_L)) * 180 / %pi;


// Standing Wave Ratio
SWR = (1 + Gamma_mag) / (1 - Gamma_mag);


// Input Impedance at distance d = 0.4λ
```

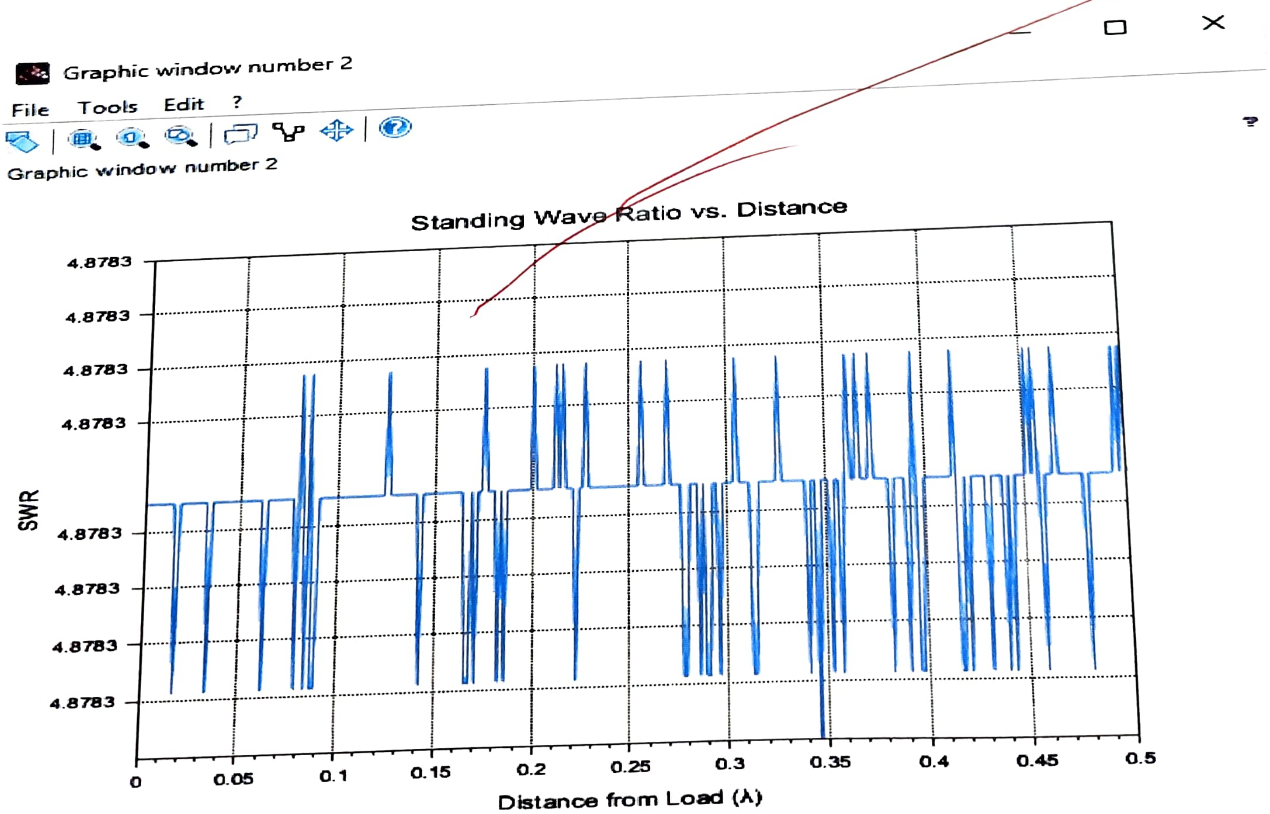
```
beta = 2 * %pi / lambda;            // Phase constant (rad/m)
Zin = Z0 * (ZL + %i*Z0 * tan(beta*d)) / (Z0 + %i*ZL * tan(beta*d));

// Display results
disp("------------------------------------------------");
disp("Reflection Coefficient (Γ_L): " + string(Gamma_L));
disp("|Γ_L| : " + string(Gamma_mag));
disp("Phase of Γ_L (degrees): " + string(Gamma_phase_deg));
disp("Standing Wave Ratio (SWR): " + string(SWR));
disp("Input Impedance at 0.4λ (Zin): " + string(Zin));
disp("------------------------------------------------");

// For visualization, sweep along 0 to 0.5λ for Γ and SWR variation
d_values = linspace(0, 0.5*lambda, 300);
Zin_values = Z0 * (ZL + %i*Z0 .* tan(beta .* d_values)) ./ (Z0 + %i*ZL .*
tan(beta .* d_values));
Gamma_d = (Zin_values - Z0) ./ (Zin_values + Z0);

// Plot |Γ| vs. distance
figure(1);
plot(d_values, abs(Gamma_d));
xlabel("Distance from Load (λ)");
ylabel("|Γ|");
title("Reflection Coefficient Magnitude vs. Distance");
xgrid();

// Plot SWR vs. distance
SWR_d = (1 + abs(Gamma_d)) ./ (1 - abs(Gamma_d));
```

```
figure(2);
plot(d_values, SWR_d);
xlabel("Distance from Load (λ)");
ylabel("SWR");
title("Standing Wave Ratio vs. Distance");
xgrid();

// Smith Chart (Γ plane)
theta = 0:0.01:2*%pi;
cx = cos(theta);
cy = sin(theta);
realG = real(Gamma_d);
imagG = imag(Gamma_d);

figure(3);
plot(cx, cy, 'k'); // unit circle
plot(realG, imagG, 'r-');
xlabel("Re(Γ)");
ylabel("Im(Γ)");
title("Smith Chart (Reflection Coefficient Plane)");
xgrid();
a = gca();
a.isoview = "on";

// Mark important points
plot(real(Gamma_L), imag(Gamma_L), 'bo');
Gamma_input = (Zin - Z0) / (Zin + Z0);
```

```
plot(real(Gamma_input), imag(Gamma_input), 'gs');
xstring(real(Gamma_L)+0.03, imag(Gamma_L), "Load (d=0)");
xstring(real(Gamma_input)+0.03, imag(Gamma_input), "Input (d=0.4λ)");
legend(["Unit circle", "Γ trajectory", "Load", "Input"], 1);
```

Output:

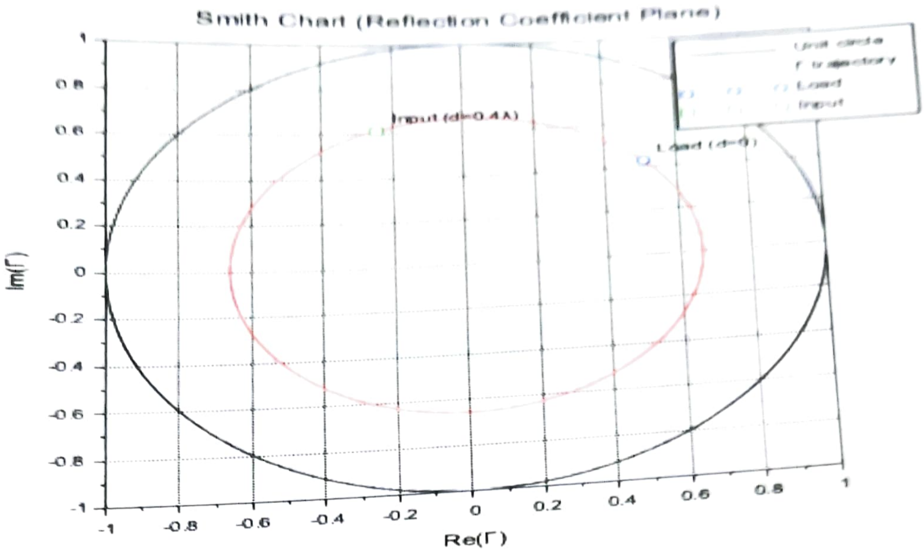




Graphic window number 3

File Tools Edit ?

Graphic window number 3



Scilab 2026.0.0 Console

File Edit Control Applications ?

Scilab 2026.0.0 Console

```
"--------------------------------------------------------"
"Reflection Coefficient (Γ_L): 0.5058824+%i*0.4235294"
"|Γ_L| : 0.6597682"
"Phase of Γ_L (degrees): 39.936383"
"Standing Wave Ratio (SWR): 4.8783458"
"Input Impedance at 0.4λ (Zin): 21.964531+%i*47.60816"
"--------------------------------------------------------"

-->
```